

q5

September 25, 2026

```
[1]: import numpy as np
import matplotlib.pyplot as plt
import scipy.stats as stats
```

```
[2]: n = 1000000
sh_analysis = stats.expon.rvs(scale=10, size=n)
extr = stats.norm.rvs(loc=15, scale=3, size=n)
eda = stats.norm.rvs(loc=3, scale=0.5, size=n)
modl = stats.poisson.rvs(mu=30, size=n)
process_int = stats.norm.rvs(loc=50, scale=20, size=n)
pipe = stats.poisson.rvs(mu=100, size=n)
modl_prod = stats.norm.rvs(loc=30, scale=5, size=n)
```

```
[3]: # Without Stakeholder analysis we cannot do development + pipeline
# No steps with EDA or after without extraction
devel = eda + modl + modl_prod
prod = pipe
print((devel > prod).sum()/n)
# percent where development takes longer than production
```

0.001286

1.

```
[4]: days = sh_analysis + extr + np.amax(np.column_stack((devel,prod)),1) +
    ↪ process_int

print("Mean number of days for process:", days.mean())
print("Median number of days for process:", np.median(days))
```

Mean number of days for process: 175.01806446150718

Median number of days for process: 174.51630290967773

```
[5]: print("two sided confidence interval:", np.quantile(days, [0.025, 0.975]))
```

two sided confidence interval: [128.04939034 224.92641197]

2.

```
[6]: print("Variance of production days:", np.var(prod))
      print("Mean of production days:", np.mean(prod))
      print("Variance of development days:", np.var(devel))
      print("Mean of development days:", np.mean(devel))
      print("Variance of process integration days:", np.var(process_int))
      print("Mean of process integration days:", process_int.mean())
      print("Variance of stakeholder analysis days:", np.var(sh_analysis))
      print("Mean of stakeholder analysis days:", sh_analysis.mean())
      print("Variance of extraction days:", np.var(extr))
      print("Mean of extraction days:", extr.mean())
```

```
Variance of production days: 99.96511678476399
Mean of production days: 99.995394
Variance of development days: 55.223406004007536
Mean of development days: 62.99018234378345
Variance of process integration days: 399.6367302622432
Mean of process integration days: 50.02606868266868
Variance of stakeholder analysis days: 99.91415811677727
Mean of stakeholder analysis days: 9.991634460462008
Variance of extraction days: 8.983715460453245
Mean of extraction days: 15.000476505592223
```

The variance is the highest for the Process Integration step, which can only be done after the Production Pipeline and model productionization are complete. Since it has the highest variance in days units, it contributes the most towards process time uncertainty.

The data pipeline contributes most to the overall duration because its mean is 100 days and is almost always the bottleneck in the dev vs prod step.

```
[7]: # So days represents our 1 million simulations of the total days for the
      ↪ process without delay
      additional_pipe_time = stats.poisson.rvs(50,size=n)
      prod_with_delay = pipe + additional_pipe_time

      days_w_delay = sh_analysis + extr + np.amax(np.
      ↪ column_stack((devel,prod_with_delay)),1) + process_int

      length_of_delay = (days_w_delay - days)
      print("Mean length of delay:", length_of_delay.mean())
      print("two sided confidence interval for length of delay:", np.
      ↪ quantile(length_of_delay, [0.025, 0.975]))
```

```
Mean length of delay: 50.00037118721575
two sided confidence interval for length of delay: [37. 64.]
```